

Image Basics

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1 Image Acquisition

- Image Acquisition using visible light on film
- Image Acquisition using visible light with digital camera
- Image Acquisition using X-ray on film

2 Image Display

3 Acknowledgements

- ▶ Image can be acquired by sensing electromagnetic waves or sound waves

- ▶ Imaging Modalities

- Light -Optical Image
 - ▶ Analog
 - ▶ Digital
- X-ray -X-ray and CT images
 - ▶ Analog X-ray Image
 - ▶ Digital X-ray Image
 - ▶ CT Images -Digital
- Gamma Ray -Gamma Image
- IR -Thermal Images

- Radio Wave -MRI images, Satellite Images(SAR)
- Micro Wave -To detect hidden objects
- Ultra sound -Ultra Sound Image

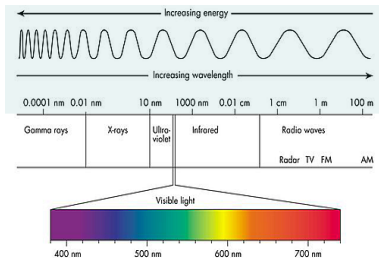


Image Acquisition using visible light on film



- ▶ Human eye is an analog camera.
- ▶ The process of computing image in human eye is incorporated in pin-hole camera.
- ▶ The image screen in pin-hole camera is a film which will get damaged when optical wave (light) hits the film.
- ▶ The damage is proportionate to the strength of light that is hitting at a position in film.

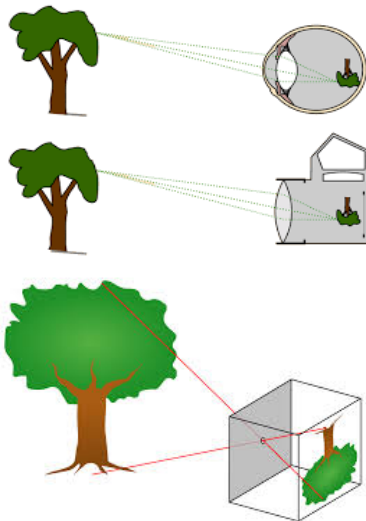


Image Acquisition using visible light on film (cont.)



- ▶ When the film is processed the complement of image is obtained in the film.
- ▶ As there is no reason why there should be gap between any two image points in the film, the image stored in film is analog
- ▶ Issue with Pin-hole camera: image is dark
 - Soln: Use lens

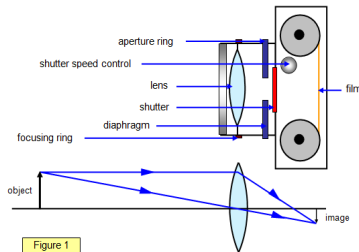
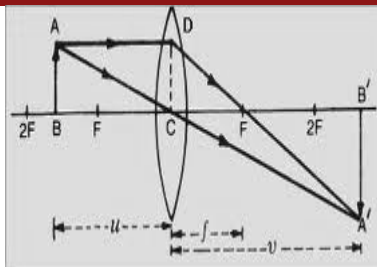


Image Acquisition using visible light on film (cont.)



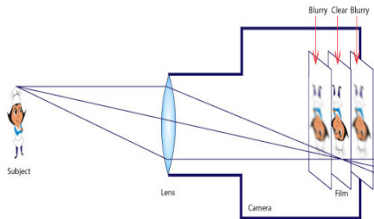
► When will good quality image be formed?

- When the lens formula:
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
 is satisfied
- Where u is the distance between object and lens, v is the distance between lens and image screen



► Magnification Formula

- $$h_i/h_o = v/u$$
- Where h_i and h_o are the heights of image and object respectively



- ▶ Digital camera uses 2D-array of sensors.
- ▶ The sensors are CCD (charge coupled device) or CMOS (Complementary Metal Oxide Semiconductor).
- ▶ Both CMOS and CCD are semiconductors, converting light signal to voltage signal.
- ▶ CMOS is required to build

high speed camera(To image sports events or high speed vehicle).

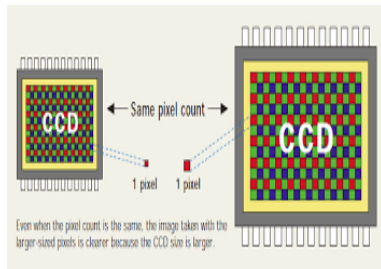
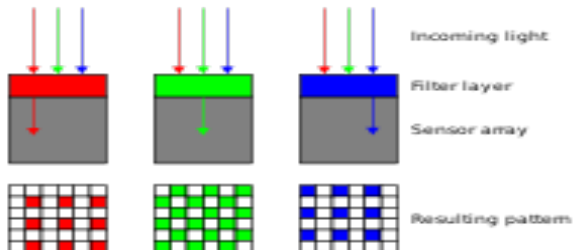


Image Acquisition using visible light with digital camera (cont.)



- ▶ Each array cell is a photo-diode, converting light intensity to voltage
- ▶ In front of each cell, there is a colour filter(RGB)
- ▶ intensity of each colour is converted to voltage





- ▶ Digital image is a 2D-array of values, called pixel values.
- ▶ Let the pixel location be (x, y) and the pixel value at the location be $f(x, y)$.
- ▶ The value of $f(x, y)$ is dependent on two factors, namely,
 - The illumination from light source that falls at point in object corresponding to (x, y)
 - The amount of reflectivity at the point in object corresponding to (x, y) .
- ▶ Let us denote the reflectivity by $r(x, y)$ and illumination by $i(x, y)$.
- ▶ The range of $r(x, y)$ be $[0, 1]$.
- ▶ Zero reflectivity means black object, and one reflectivity means white object.
- ▶ The reflectivity of black velvet is 0.01 and of snow is 0.93.



- ▶ As a portion of amount of light which is illuminated is reflected, and the same is captured by the sensor
- ▶ The **Optical Imaging model**: $f(x, y) = r(x, y) * i(x, y)$. Using this model given any two of f , r and i , we can find the other one
- ▶ Defn of Digital gray scale Image: Matrix of size $m \times n$, with integers 0 to $L - 1$
- ▶ Defn of Digital colour image Image: Triplet of Matrices (R, G, B) of size $m \times n$ each, with integers 0 to $L - 1$, where
 - All three matrices are gray scale images
 - R matrix represents the red colour of the received white light
 - G matrix represents the green colour of the received white light
 - B matrix represents the blue colour of the received white light

- ▶ The film based X-ray imaging has a film which will get damaged if X-ray hits the film.
- ▶ The amount of damage is proportionate to the intensity of X-ray hit at the point.
- ▶ An object to be imaged will be kept between the X-ray source and the X-ray film.
- ▶ The X-ray emitted by the

object will penetrate through the object and hit the X-ray film.

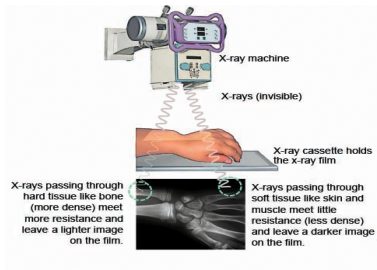


Image Acquisition using X-ray on film (cont.)



- ▶ The intensity of X-ray penetrate through the object is dependent on the hardness of the object along the line.
- ▶ After exposing the film to X-ray, the film will be processed to get the image complement(intensity black to white and white black).
- ▶ In this process, there is no reason why a particular point in X-ray film is not damaged,
- ▶ Amplitude is also non-discrete. Hence the film based X-ray image is analog image.



► Monitors/TV

- CRT, LCD, LED, OLED, Plasma

► Printers

- Dot matrix, inkjet, laser

► Projectors



MATRIX PRINTER

vs.



INKJET PRINTER



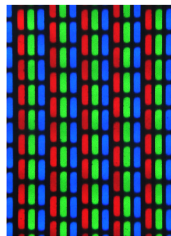
Image Display (cont.)



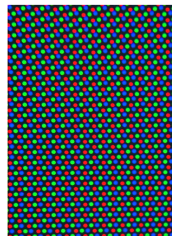
- ▶ CRT uses phosphorous coated screen
- ▶ The screen is divided into grid points
- ▶ Each grid point will have three tiny dots, where

which glows in BLUE colour when electron hits the dot

- The first dot is coated with one type of phosphorous which glows in RED colour when electron hits the dot
- The third dot is coated with one type of phosphorous which glows in GREEN colour when electron hits the dot
- The third dot is coated with one type of phosphorous

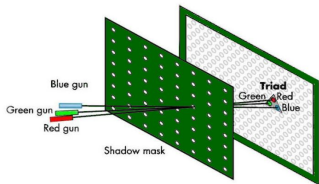


21" TV CRT Display



17" PC CRT Display

Colour CRT



Shadow mask techniques

- Images are downloaded from internet sources

Thank You! :)